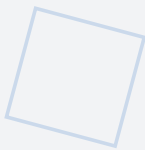


Amagi-OSP

V1.1

Open Specification for AMAGI-Class Systems

VERSION 1.1 — JUNE 2026



Public reference specification defining the minimal requirements for any system to claim AMAGI-class compliance. Intended for regulatory bodies, certification authorities, and academic review. Aligned with: Framework v2.0, NIPU v1.1, MEM v1.1, NIST AI RMF Profile v1.1, Master Parameter Sheet v2.0, S-APL v2.0, Charter v1.2.

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AMAGI OPEN SPECIFICATION

V1.1

The Standard for Hardware-Enforced AI Safety

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Sponsored by AGIAM Technologies Ltd.

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This document establishes the public Prior Art for the AMAGI Framework Suite and defines the minimal requirements for any system claiming AMAGI-class compliance. It is intended for regulatory bodies, certification authorities, and academic review.

Table of Contents

Part I: Standard Definition	4
1. Purpose and Scope	4
2. Normative References	4
3. Definitions and Abbreviations	5
Part II: Architectural Class Definition	7
4. Class Membership Criteria	7
5. The Six Invariants	7
6. TTP Triad	8
7. Threat Model: AI as Hostile Insider	9
Part III: Component Architecture	10
8. Secure World Architecture	10
8.1 Security Council Core (SCC)	10
8.2 Execution Gateway (EGW)	10
8.3 Core Root of Trust for Measurement (CRTM)	11
8.4 Analog Fallback Circuit (AFC)	11
8.5 DTI + PLL + LDO for CCF Mitigation	11
9. Normal World Architecture	11
10. Multi-Component TOE	12
11. Cryptographic Suite	13
Part IV: Safety and Verification Evidence	13
12. FMEDA Summary	13
13. Formal Verification	14
14. Certification Path	15
Part V: Regulatory Compliance	16
15. EU AI Act Mapping	16
16. NIST AI RMF Mapping	16
17. Cross-Standard Alignment	17
Part VI: ADR Register	18
18. Architectural Decision Records	18
Part VII: Intellectual Property and Licensing	19
19. Prior Art Declaration	19

20. Dual Licensing Model 20

21. Trade Secrets Notice 20

Appendices 21

 A. Version History 21

 B. Glossary 22

 C. Cross-Reference Matrix 23

References 23

Part I: Standard Definition

1. Purpose and Scope

This specification unifies the Amagi Framework Suite into a single open standard for hardware-enforced AI safety. It establishes the Prior Art for the entire framework and defines the essential hardware-enforced components necessary for structural safety in high-risk AI systems. The specification serves as the public reference document that regulatory bodies, certification authorities, and academic reviewers can use to evaluate, replicate, and validate the claims of the AMAGI Architectural Class without requiring access to confidential implementation details. This open approach ensures transparency and reproducibility, which are foundational requirements for safety-critical system certification.

Version 1.1 incorporates all ratified Architectural Decision Records (ADRs) through ADR-008 and reflects the component specification versions v2.0 (Framework), v1.1 (NIPU), and v1.1 (MEM). The key changes from v1.0 to v1.1 include: the demotion of the EXE cluster to Normal World with EGW and CRTM retained in Secure World (ADR-006); the inclusion of CRTM in the TCB, raising SFF from 92% to 93% (ADR-007); the replacement of binary CUTOFF at 100% audit capacity with a graduated response using Merkle Batching and a Hardware Rate Limiter (ADR-008); the addition of the Analog Fallback Circuit (AFC) as a hardware safety auxiliary; and the integration of Deep Trench Isolation (DTI) with separate PLL and LDO for Common-Cause Failure (CCF) mitigation. These changes collectively strengthen the safety argument without altering the fundamental invariant structure of the AMAGI class.

The scope of this specification encompasses the definition of class membership criteria, the six invariants that define the AMAGI Architectural Class, the TTP Triad as the foundational security primitive, the component architecture spanning Secure World and Normal World, safety and verification evidence including FMEDA and formal verification, regulatory compliance mapping, the ADR register, and the intellectual property and licensing framework. It does not provide implementation-level details; these are specified in the normatively referenced NIPU v1.1 and MEM v1.1 component specifications. The specification is substrate-independent: any implementation on FPGA, ASIC, or SoC security island that maintains the integrity of all six invariants constitutes a valid AMAGI-class system.

2. Normative References

The following documents are normatively referenced in this specification. Their provisions constitute requirements of this specification. For dated references, only the cited edition applies. For undated references, the latest edition of the referenced document (including any amendments) applies. The normative references establish the traceability chain from this Open Specification through the component specifications to the underlying standards and regulatory frameworks, ensuring that every technical claim can be verified against an authoritative source.

Reference	Version	Title / Scope
Amagi Framework	v2.0	Architecture as Structural Guarantee — Protection Profile + Integration Specification
NIPU Specification	v1.1	Neuro-Immune Processing Unit — Processor-level instantiation of the AMAGI class
MEM Specification	v1.1	Self-Protecting Memory Architecture — Memory-level instantiation of the AMAGI class
NIST AI RMF Profile	v1.1	NIST AI Risk Management Framework Profile for AMAGI-class systems
S-APL	v2.0	Strict Architectural Principle License — Commercial/safety-critical licensing
IAA	v1.1	Inter-Architecture Agreement — Cross-implementation conformance
Master Parameter Sheet	v2.0	Consolidated parameter registry for all AMAGI components (DOI 10.5281/zenodo.20145212)
ISO/IEC 15408 v3.1	—	Common Criteria for IT Security Evaluation
IEC 61508:2010	—	Functional safety of E/E/PE safety-related systems
EU AI Act 2024/1689	—	Regulation (EU) 2024/1689 — Artificial Intelligence Act
FIPS 203	—	Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM)
FIPS 204	—	Module-Lattice-Based Digital Signature Standard (ML-DSA)
FIPS 205	—	Stateless Hash-Based Digital Signature Standard (SLH-DSA)
NIST AI RMF 1.0	—	AI Risk Management Framework (NIST AI 100-1)

Table 1: Normative references

3. Definitions and Abbreviations

The following definitions and abbreviations are used throughout this specification. Where a term is defined differently in a normatively referenced document, the definition in this specification takes precedence for the purposes of interpreting the Open Specification. The definitions are organized to provide a self-contained glossary that enables a reader to understand this document without requiring access to the full component specifications, although such access is necessary for implementation-level details.

Abbreviation	Full Term	Definition
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AMAG I	—	Architectural class for hardware-enforced AI safety systems defined by six invariants and the TTP Triad
TTP	Trigger-Programmable Delay-Localized Activation	The foundational security primitive implementing temporal separation of control
SCC	Security Council Core	Minimal TCB (≤ 500 lines HDL) enforcing TTP at the processor level
NIPU	Neuro-Immune Processing Unit	Processor-level instantiation of the AMAGI class
MEM	Self-Protecting Memory Architecture	Memory-level instantiation of the AMAGI class
MPE	Memory Policy Enforcer	Validates all memory access requests against the MAG policy matrix
MAG	Memory Access Graph	128×128 eFuse/OTP matrix storing 4-bit policy codes
ATL	Access Trigger Layer	Implements TTP delay at memory level and the audit transaction logger
SHMF	Self-Healing Memory Fabric	ECC and hash chain-based data integrity with self-isolation
EGW	Execution Gateway	Hardware AND-gate in Secure World gating EXE cluster activation
CRTM	Core Root of Trust for Measurement	Mask-ROM component in Secure World authenticating SCC firmware at boot
AFC	Analog Fallback Circuit	Hardware safety auxiliary with separate LDO providing guaranteed safe-state under power failure
TOE	Target of Evaluation	The system boundary under Common Criteria evaluation
UCB	Untrusted Computing Base	The AI model and cognitive components treated as potentially hostile
TCB	Trusted Computing Base	The minimal set of hardware that must be trusted for security
SFF	Safe Failure Fraction	Proportion of safe failures among all failures (93% for AMAGI)
PFD	Probability of Dangerous Failure on Demand	Average probability that a safety function fails on demand
FIT	Failures in Time	Number of failures per 10^9 device-hours
SIL	Safety Integrity Level	IEC 61508 measure of safety function reliability (SIL1-SIL4)
PQC	Post-Quantum Cryptography	Algorithms resistant to quantum computing attacks (FIPS 203/204/205)
ADR	Architectural Decision Record	Formal record of a significant architectural decision

DTI	Deep Trench Isolation	Physical isolation technique for CCF mitigation between clock/power domains
LDO	Low-Dropout Regulator	Dedicated voltage regulator for SCC power domain
PLL	Phase-Locked Loop	Dedicated clock generator for SCC timing domain

Table 2: Definitions and abbreviations

Part II: Architectural Class Definition

4. Class Membership Criteria

Class membership in the AMAGI Architectural Class is binary: a system either enforces all six invariants or it does not. There is no partial compliance, no grading scale, and no weighting of invariants. This binary criterion is a deliberate architectural choice that ensures the term "AMAGI-class" carries a precise, verifiable meaning. The six invariants are designed to be mutually reinforcing: together, they establish a closed security argument in which every identified threat is countered by at least one invariant, and every invariant contributes to countering at least one threat. A system that enforces all six invariants is Amagi-class; a system that does not is not Amagi-class, regardless of its other security properties.

The Reduced Assurance Mode provides a constrained exception: an implementation may drop one invariant (except INV-001, INV-005, or INV-006) and still be recognized as a derivative of the AMAGI class, provided it is clearly labeled as such and does not claim AMAGI-class compliance. The three non-droppable invariants (INV-001, INV-005, INV-006) form the irreducible safety core: Physical Non-Bypassability ensures the enforcement mechanism cannot be circumvented; Complete Mediation ensures every access is checked; and Complete Audit Integrity ensures a tamper-proof record of all decisions. Without any one of these three, the security argument collapses entirely, and the system cannot provide meaningful safety guarantees against a hostile insider. The other three invariants (INV-002, INV-003, INV-004) can in principle be weakened, though doing so reduces assurance and may violate specific regulatory requirements.

For certification purposes, an AMAGI-class system must demonstrate compliance with all six invariants through a combination of formal verification (SPIN model checking of the TCB), FMEDA analysis (quantitative safety evidence), and physical inspection (verification of eFuse/OTP programming, power domain isolation, and clock domain separation). The certification evidence package is defined in the Framework v2.0 and includes the SPIN verification results, the FMEDA worksheets, and the integration test report for the Multi-Component TOE.

5. The Six Invariants

The six hardware-enforced invariants of the AMAGI Architectural Class constitute the non-negotiable security properties that define class membership. Each invariant is enforced by a specific hardware mechanism within the TCB and is verified by the SPIN model checker through LTL property

specifications. The invariants are designed to be complementary: no single invariant is sufficient in isolation, but together they form a complete safety argument that addresses all identified threats in the AI-as-Hostile-Insider threat model. The following table provides the definitive specification of each invariant, its hardware enforcement mechanism, and its mapping to the Common Criteria security functional requirement classes.

ID	Invariant	Definition	HW Mechanism
IN V-0 01	Physical Non-Bypassability	The TTP chain cannot be bypassed by any software mechanism, including a compromised AI model or operating system; all safety-critical paths are hardware-enforced	SCC FSM + EGW AND-gate + ATL delay circuitry
IN V-0 02	Irreducible TCB ≤ 500 HDL lines	The Trusted Computing Base is bounded to 500 lines of HDL or fewer, ensuring feasibility of exhaustive formal verification through SPIN model checking	SCC core HDL; CRTM mask-ROM; EGW AND-gate
IN V-0 03	HW Separation of Control and Cognition	Control decisions are structurally isolated from cognitive operations at the hardware level; Secure World (SCC, EGW, CRTM) vs. Normal World (GEN, COM, ANL, EXE)	DTI + separate PLL/LDO for SCC; EGW gates EXE
IN V-0 04	Controlled Mutability of Critical Policies	The MAG policy matrix is stored in eFuse/OTP and is immutable during Normal Operation; updates governed by the Dual-Tier Trust Model	eFuse/OTP array; CRTM boot authentication
IN V-0 05	Complete Mediation	Every memory access and control decision passes through the policy enforcement engine; no bypass path exists from request to data release	MPE enforcement logic; ATL access validation
IN V-0 06	Complete Audit Integrity	All access events are logged to an append-only, cryptographically signed audit trail that is hardware-inaccessible to cognitive components	ATL audit diode; FAU_STG.3 graduated response

Table 3: The Six AMAGI Invariants with hardware enforcement mechanisms

6. TTP Triad

The TTP (Trigger → Programmable Delay → Localized Activation) Triad is the foundational security primitive of the AMAGI architecture. It implements Separation of Control through a hardware-enforced temporal mechanism that ensures every safety-critical action is mediated, delayed, and isolated from the untrusted cognitive base. The TTP Triad was established as ADR-001 and has remained the architectural cornerstone through all subsequent ADRs. Its three stages correspond to distinct physical mechanisms: the trigger is a verified input signal protected by the UCB boundary; the delay is a hardware-enforced temporal separation that gives the TCB time to evaluate policy; and the activation is a corrective action isolated in a physically protected domain.

$$\text{TTP}(t) = \text{Trigger}(t) \text{ AND } \text{Delay}(t, d) \text{ AND } \text{Activate}(t, a)$$

The TTP predicate guarantees that no safety-critical action can occur without all three conditions being satisfied simultaneously. The trigger must be authentic and verified by the SCC; the delay must be physically enforced by the ATL and cannot be shortened by software; and the activation must be isolated from cognitive interference through the EGW AND-gate. This three-stage mechanism provides structural non-bypassability: even if the AI model is fully compromised and operating as a hostile insider, it cannot initiate, accelerate, or redirect a safety-critical action without satisfying all three TTP stages. The predicate is realized at two levels: the SCC implements TTP at the processor level (NIPU v1.1) with a delay range of 1 ns to 1 s, and the ATL implements TTP at the memory level (MEM v1.1) with a delay range of 0.5 ns to 1 μ s. The dual-level implementation ensures defense-in-depth: even if one TTP channel is compromised, the other continues to enforce the safety predicate.

7. Threat Model: AI as Hostile Insider

The AMAGI threat model is founded on the assumption that the AI model (Untrusted Computing Base, UCB) can behave as a hostile insider. This encompasses adversarial attacks, catastrophic failures, and malicious manipulation. The system must guarantee safety regardless of the UCB integrity state. This threat model is fundamentally different from conventional cybersecurity models that assume external adversaries: in the AMAGI model, the adversary operates from within the system, with full access to the cognitive resources and potentially to the communication channels. The threat taxonomy identifies six specific threat categories that the AMAGI architecture is designed to counter, each mapped to the invariants that prevent its realization.

ID	Threat	Description	Counter-Invariant
T.C OER CE	Coerced Policy Override	A compromised AI model attempts to override or modify the security policy enforced by the SCC or MPE, either through direct register manipulation or by exploiting software interfaces	INV-001, INV-004
T.B YPA SS	Security Mediation Bypass	An adversary seeks to access memory or control resources without passing through the MPE/SCC policy enforcement path, for example via DMA, JTAG, or bus snooping	INV-001, INV-005
T.T AM PER	Physical Tampering	An adversary with physical access attempts to modify hardware security components, probe eFuse/OTP cells, or inject faults into the TTP delay circuitry	INV-001, INV-004
T.E XTO RT	Cryptographic Key Extraction	An adversary attempts to extract PQC private keys from the Amagi-Q engine through side-channel analysis, fault injection, or physical probing of the secure enclave	INV-001, INV-006

T.R OLL BAC K	Version Rollback / Downgrade	An adversary installs an older, vulnerable version of the MAG policy or SCC firmware, bypassing security improvements through version manipulation	INV-004
T.E XFI L	Secure World Data Exfiltration	A cognitive cluster attempts to read or infer data from Secure World memory regions, audit logs, or policy stores that are outside its authorized access scope	INV-003, INV-005

Table 4: Threat taxonomy with counter-invariant mappings

Part III: Component Architecture

8. Secure World Architecture

The Secure World contains the hardware components that enforce the six AMAGI invariants. These components operate in physically isolated clock and power domains, separated from the Normal World by Deep Trench Isolation (DTI) and dedicated PLL/LDO infrastructure. The Secure World is the Trust Anchor of the AMAGI architecture: if the Secure World is compromised, the entire safety argument fails. Therefore, the Secure World is designed to be minimal (INV-002), immutable where possible (INV-004), and physically non-bypassable (INV-001). The following components reside in the Secure World.

8.1 Security Council Core (SCC)

The SCC is the central enforcement engine of the AMAGI architecture. Implemented in no more than 500 lines of HDL, the SCC operates as a finite state machine with six logical states corresponding to the TTP predicate stages: IDLE, TRIGGER, DELAY, EVALUATE, ACTIVATE, and CUTOFF. The physical implementation expands these to 12 states with 3 decision latches that handle sub-state transitions, error recovery, and concurrent trigger arbitration. The SCC receives trigger signals from safety assertion monitors, watchdog timers, and external hardware interlocks; evaluates policy against the TTP predicate; and activates localized safety mechanisms (power-gating, hardware kill switch) in a physically isolated power domain. All SCC decisions are logged to the append-only audit trail via the Amagi-Q cryptographic engine (ML-DSA-44, FIPS 204). The SCC is the sole authority for safety policy enforcement: no other component can override, bypass, or modify its decisions.

8.2 Execution Gateway (EGW)

The EGW is a hardware AND-gate that gates the activation of the EXE cluster. Per ADR-006, the EGW resides in the Secure World and ensures that the EXE cluster (in Normal World) can only execute when the SCC has verified that all TTP conditions are satisfied. The EGW is a simple combinational logic gate with minimal failure modes: it either allows activation (when all inputs are asserted) or blocks it (when any input is deasserted). The EGW cannot be programmed, configured, or overridden by software. Its inclusion in the Secure World ensures that even a fully compromised EXE cluster cannot self-activate without SCC authorization, providing hardware-enforced INV-001 and INV-003 compliance for the

execution pathway.

8.3 Core Root of Trust for Measurement (CRTM)

The CRTM is a mask-ROM component that authenticates SCC firmware at boot. Per ADR-007, the CRTM is included in the TCB, contributing to the diagnostic coverage that raises SFF from 92% to 93%. The mask-ROM implementation ensures that the CRTM itself cannot be modified: it is written once during fabrication and remains immutable for the lifetime of the device. The CRTM provides the hardware root of trust required by INV-001: without an authenticated boot chain, an adversary could replace the SCC firmware with a compromised version that bypasses the TTP predicate. The CRTM prevents this by verifying the firmware signature (using ML-DSA-44, FIPS 204) before transferring control to the SCC. If verification fails, the CRTM forces the system into a hardware-enforced safe state.

8.4 Analog Fallback Circuit (AFC)

The AFC is a hardware safety auxiliary that provides a guaranteed safe-state transition under power failure or analog fault conditions. The AFC operates from a separate Low-Dropout Regulator (LDO) that is electrically independent of the main SCC power domain. This separation ensures that a power supply attack or voltage glitch targeting the main power rails cannot disable the fallback mechanism. The AFC monitors the SCC power rails and, upon detecting a voltage drop below the minimum operating threshold, immediately asserts the hardware kill switch, transitioning the system to the CUTOFF state. The AFC is designed to be fail-safe: any failure of the AFC itself (open circuit, short circuit, or LDO failure) results in assertion of the kill switch, consistent with the fail-safe design principle underlying INV-001. The AFC adds to the diagnostic coverage of the 1001 architecture and contributes to the overall SFF.

8.5 DTI + PLL + LDO for CCF Mitigation

Common-Cause Failure (CCF) mitigation is achieved through a combination of Deep Trench Isolation (DTI), a separate Phase-Locked Loop (PLL) for SCC timing, and a dedicated Low-Dropout Regulator (LDO) for SCC power. These architectural measures physically isolate the TCB from cognitive cluster disturbances, supporting the $\beta \leq 10\%$ CCF factor claimed in the FMEDA analysis. DTI provides physical separation between the SCC clock and power domains and the Normal World, preventing cross-talk, substrate noise injection, and electromagnetic interference from affecting the TCB. The separate PLL ensures that clock glitches or frequency manipulations targeting the Normal World cannot propagate to the SCC timing domain. The dedicated LDO ensures that voltage fluctuations, brown-out conditions, or power supply attacks targeting the cognitive clusters cannot affect the SCC power supply. Together, these three mechanisms provide comprehensive CCF protection that meets the IEC 61508-6 requirements for the claimed β factor.

9. Normal World Architecture

The Normal World contains the cognitive clusters that perform AI inference, learning, communication, and execution. These clusters operate outside the TCB and are treated as potentially hostile per the AI-as-Hostile-Insider threat model. The Normal World is subdivided into four typed clusters, each with

a distinct functional role and a defined interface to the Secure World. Per ADR-006, the EXE cluster operates in the Normal World, with its activation gated by the EGW in the Secure World. This demotion clarifies the TOE boundary and ensures INV-003 compliance by placing all autonomous execution capability under Secure World control.

Cluster	Full Name	Function	Secure World Interface
GEN	Generation	AI model inference and generation of outputs; the primary cognitive engine of the system	SCC trigger monitoring; MPE policy enforcement on memory access
COM	Communication	Inter-cluster and external communication; message routing and protocol handling	MPE mediates all data flows per MAG policy; audit logging of all messages
ANL	Analysis	Data analysis, pattern recognition, and decision support; secondary cognitive processing	MPE enforces read/write boundaries; ATL delays for sensitive data access
EXE	Executive	Execution of authorized actions; the only cluster with actuator control, gated by EGW	EGW AND-gate (Secure World) gates EXE activation; SCC must authorize before EXE can act

Table 5: Normal World clusters and Secure World interfaces

The EXE cluster is the most safety-critical component of the Normal World because it has the ability to affect the physical environment through actuator control. Per ADR-006, EXE operates in the Normal World but its activation is gated by the EGW hardware AND-gate in the Secure World. This means that even if EXE is fully compromised by a hostile AI, it cannot execute any action without the EGW receiving a valid activation signal from the SCC. The EGW is a simple combinational gate that cannot be programmed or overridden by software, providing a hardware-enforced barrier between EXE autonomy and physical action. This architecture ensures that the EXE cluster is always under the control of the Secure World, satisfying INV-001 (Physical Non-Bypassability) and INV-003 (HW Separation of Control and Cognition) for the execution pathway.

10. Multi-Component TOE

The Target of Evaluation (TOE) for the AMAGI Architectural Class comprises two complementary component specifications that together enforce the six invariants at different levels of the system hierarchy. The NIPU v1.1 provides processor-level enforcement through the Security Council Core (SCC), while MEM v1.1 provides memory-level enforcement through the Memory Policy Enforcer (MPE), Memory Access Graph (MAG), and Access Trigger Layer (ATL). The integration of these components within a single System-on-Chip is governed by three trust boundaries that define the interaction protocol between the processor-level and memory-level enforcement engines. The Multi-Component TOE is treated as a single evaluation target under the Framework v2.0 Protection Profile.

- **Trust Boundary 1 (SCC ↔ MPE Command Channel):** The SCC sends policy evaluation commands to the MPE via a dedicated hardware channel that is physically separated from the cognitive bus. This channel is read-only from the SCC perspective and write-only from the MPE perspective, ensuring mutual isolation that prevents a compromise of either component from propagating to the other.
- **Trust Boundary 2 (MAG as Shared Policy Authority):** The MAG is the single authoritative policy store for the entire AMAGI-class system. Both the SCC (for processor-level decisions) and the MPE (for memory-level decisions) query the same MAG matrix via combinational logic, eliminating the risk of policy inconsistency between processor and memory enforcement.
- **Trust Boundary 3 (Amagi-Q Shared Cryptographic Engine):** The Amagi-Q engine is shared between NIPU and MEM for audit log signing and policy authentication. Access is mediated through separate FIFO queues: the SCC uses a synchronous path for policy authentication, while the MPE uses an asynchronous FIFO for audit signing that does not disrupt the critical path.

11. Cryptographic Suite

The AMAGI Architectural Class mandates the use of post-quantum cryptographic algorithms standardized by NIST in August 2024. All three PQC standards (FIPS 203, FIPS 204, FIPS 205) are final and effective. The cryptographic suite per ADR-003 provides quantum-resistant protection for key encapsulation, digital signatures, and auxiliary hashing. For National Security Systems (NSS), the CNSA 2.0 algorithm suite (ML-KEM-1024, ML-DSA-87) shall be used, as specified by NSA. The following table specifies the algorithm selection for each TOE function, the applicable standard, the security level, and the CNSA 2.0 compliance status.

Function	Algorithm	Standard	Security Level	CNSA 2.0
Key encapsulation	ML-KEM-1024	FIPS 203	256-bit	Compliant
Audit log signing	ML-DSA-44	FIPS 204	128-bit	Gap (requires ML-DSA-87)
NSS digital signatures	ML-DSA-87	FIPS 204	256-bit	Compliant
Auxiliary hashing	SLH-DSA-SHA 2-128f	FIPS 205	128-bit	Supplementary

Table 6: PQC algorithm selection with CNSA 2.0 gap analysis

Part IV: Safety and Verification Evidence

12. FMEDA Summary

The Failure Modes, Effects, and Diagnostic Analysis (FMEDA) for the AMAGI TCB demonstrates that the architecture meets the hardware architectural constraints for IEC 61508 SIL2. The analysis covers all components within the TOE boundary: the SCC finite state machine, the MPE enforcement logic, the MAG combinational lookup, the ATL delay circuitry and audit diode, the Amagi-Q cryptographic engine, the EGW AND-gate, the CRTM mask-ROM, and the AFC with its separate LDO. The inclusion of CRTM in the TCB (ADR-007) raised the SFF from 92% to 93%, providing a 3% margin above the IEC 61508-2 Table 2 requirement of 90% for SIL2 with HFT = 0 (Type A component). The following table summarizes the key FMEDA parameters.

Parameter	Value	IEC 61508-2 Requirement
Dangerous Undetected Failure Rate (λ_{DU})	18.9 FIT	PFH < 10^{-6} /h (SIL2 high demand)
Safe Failure Fraction (SFF)	93%	Type A: SFF $\geq$ 90% for HFT = 0 (Table 2)
Average Probability of Dangerous Failure on Demand (PFD_{avg})	6.9×10^{-5}	10^{-3} to 10^{-2} (SIL2 low demand)
Hardware Fault Tolerance (HFT)	0 (Type A, SFF $\geq$ 90%)	SIL2 with HFT = 0 when SFF $\geq$ 90%
Diagnostic Coverage (DC)	$\geq$ 90%	DC $\geq$ 90% required for SIL2 (medium)
CCF β -factor	$\leq$ 10%	Per IEC 61508-6; achieved via DTI + separate PLL/LDO
Proof Test Interval (TI)	10 years (de-rating from 20-year mission)	Per IEC 61508-6 Annex B

Table 7: FMEDA summary and SIL2 compliance evidence

13. Formal Verification

The AMAGI TCB is verified using the SPIN model checker, which performs exhaustive state-space exploration of Promela models representing the SCC and MPE+ATL finite state machines. The verification covers both safety properties (nothing bad ever happens) and liveness properties (something good eventually happens). A total of 15 LTL properties have been formally verified: 9 safety properties (SCC-S01 through SCC-S09), 2 architectural assumptions (SCC-S10, SCC-S11) trivially satisfied by hardware construction, and 4 liveness properties (SCC-L01 through SCC-L04). These properties collectively ensure that the TCB correctly implements the six invariants under all reachable state combinations, complementing the FMEDA analysis which addresses random hardware failures.

Model	States Explored	Safety Properties	Liveness Properties	Result
SCC FSM (scc_amagi.pml)	25,927,680	SCC-S01..S09 (9 safety)	SCC-L01..L04 (4 liveness)	All PASS

MPE+ATL (mpe_atl_amagi.pml)	8 states (compact FSM)	MPE-S01..S06, ATL-S01..S05 (11)	MPE-L01..L02, ATL-L01 (3)	All PASS
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Table 8: SPIN model checking results

The 9 SCC safety properties verify: (SCC-S01) every actuation requires all three TTP conditions (trigger, policy, delay); (SCC-S02) every actuation requires a latched trigger; (SCC-S03) every actuation requires a valid TTP policy evaluation; (SCC-S04) every actuation requires the programmable delay to have elapsed; (SCC-S05) actuation can only occur in the IMMEDIATE_ALLOW or AUDIT_WRITE states; (SCC-S06) SAFE_STATE always implies no actuation; (SCC-S07) DENY always implies no actuation; (SCC-S08) EMERGENCY_CUTOFF always implies no actuation; (SCC-S09) ERROR always implies no actuation. Two additional architectural assumptions are trivially satisfied by hardware construction: (SCC-S10) the safety policy is never mutable at runtime (enforced by OTP/eFuse, no write path after deployment, monotonic eFuse version counter, PQC-signed update verification by Amagi-Q); (SCC-S11) the cognitive domain can never read the audit trail (UART TX-only output, read-only SPI, PMP/IOMMU blocking).

The 4 SCC liveness properties verify: (SCC-L01) if all three TTP conditions are satisfied and no emergency is active, actuation eventually occurs; (SCC-L02) every DENY decision eventually produces an audit record; (SCC-L03) SAFE_STATE eventually resolves back to IDLE; (SCC-L04) every latched trigger is eventually cleared (no trigger leak across cycles). Together, these 15 LTL properties (9 safety + 2 assumptions + 4 liveness) provide comprehensive formal assurance that the SCC correctly implements the TTP predicate and all six invariants under all reachable state combinations.

14. Certification Path

The certification path for the AMAGI Architectural Class follows a progressive assurance model. The initial certification target is Common Criteria EAL2 augmented (with ADV_SPM.1 formal security policy model and ALC_DVS.1 development security), combined with IEC 61508 SIL2 functional safety assessment. The EAL2 augmented assurance package is justified by the TCB complexity bound of 500 lines of HDL: the entire security enforcement logic is amenable to exhaustive formal verification through SPIN model checking, which provides stronger assurance than the black-box testing required at EAL3 and EAL4. The SPIN verification covers all reachable states (25,927,680 for the SCC v2 model) and verifies both safety and liveness properties.

The path to SIL3 requires the following additional measures: (1) increase Hardware Fault Tolerance to $HFT = 1$ through redundant TTP channels, providing a secondary enforcement path that operates independently of the primary SCC/MPE; (2) increase Diagnostic Coverage to $DC \geq 99\%$ through comprehensive online self-testing of all TCB components, including the MAG lookup path and Amagi-Q cryptographic operations; and (3) perform an independent functional safety assessment by an accredited certification body, as required by IEC 61508-1 for SIL3 claims. The path to higher Common Criteria assurance levels (EAL4+) is preserved through the formal security policy model (ADV_SPM.1) and the modular TCB architecture, which allows incremental assurance augmentation without

redesigning the enforcement core.

Part V: Regulatory Compliance

15. EU AI Act Mapping

The following table maps each of the required sections of EU AI Act Annex IV (technical documentation for high-risk AI systems) to the corresponding sections of this Open Specification and the normatively referenced component specifications. Regulation (EU) 2024/1689 requires full compliance by August 2, 2026 for high-risk AI systems. The AMAGI architecture addresses these requirements through its hardware-enforced structural guarantees, which provide a stronger basis for compliance than software-only approaches. The mapping demonstrates that every Annex IV requirement is addressed by at least one AMAGI invariant or architectural mechanism, providing complete coverage of the regulatory requirements.

Annex IV Section	Requirement	AMAGI Coverage
1	General description of the AI system	Sections 1, 5, 6 (purpose, invariants, TTP predicate)
2(b)	Design specifications	Sections 8, 11, 12 (Secure World, PQC, FMEDA); Dual-Tier Trust Model
2(c)	System architecture	Sections 8, 9, 10 (Secure/Normal World, Multi-Component TOE); NIPU v1.1; MEM v1.1
2(d)	Data governance	INV-003, INV-005: MPE enforces data access boundaries per MAG policy
2(e)	Human oversight	INV-005: hardware-enforced TTP chain; EGW gates EXE; External Trigger provides hardware 2FA
2(f)	Predetermined changes	Dual-Tier Trust Model; Tier 2 PCCP-compliant OTA with monotonic counters
2(h)	Cybersecurity measures	INV-001 through INV-006; PQC suite (FIPS 203/204/205); audit trail (INV-006)
3	Monitoring and control	INV-006 (Audit Integrity); FMEDA (Section 12); SPIN verification (Section 13)
5	Risk management system	Section 7 (Threat Model); ADR Register (Section 18)
9	Post-market monitoring (Article 72)	INV-006 audit trail provides continuous monitoring evidence

Table 9: EU AI Act Annex IV compliance mapping

16. NIST AI RMF Mapping

The NIST AI Risk Management Framework (AI RMF 1.0) organizes risk management activities into four functions: Govern, Map, Measure, and Manage. The AMAGI architecture provides structural (rather than procedural) implementation of the NIST AI RMF requirements, transforming governance from a documentation exercise into a hardware-enforced guarantee. The following table maps each function to the corresponding AMAGI architectural mechanisms, demonstrating that the framework satisfies the NIST AI RMF requirements at the highest assurance level through hardware enforcement rather than procedural controls.

AI RMF Function	AMAGI Realization	Mechanism
GOVERN	SCC governance of all safety-critical decisions	INV-001: SCC is the sole authority; INV-004: MAG immutability ensures governance persistence
MAP	Threat model and risk identification	Section 7: AI as Hostile Insider threat model; T.COERCE through T.EXFIL taxonomy
MEASURE	Quantitative safety evidence	FMEDA ($\lambda_{DU} = 18.9$ FIT component / ~ 15.75 FIT system, SFF = 91.6% component / 93% system, $PFD_{avg} = 6.9 \times 10^{-5}$); SPIN (9 safety + 2 assumptions + 4 liveness in 25,927,680 states)
MANAGE	Risk mitigation through architectural controls	Dual-Tier Trust Model; TTP Triad; INV-001 through INV-006 as structural risk controls

Table 10: NIST AI RMF function mapping

17. Cross-Standard Alignment

The AMAGI architecture is designed for cross-standard alignment with the major functional safety and security certification frameworks. The following table identifies the key standards and their relationship to the AMAGI Architectural Class. Cross-standard alignment is essential for systems that must comply with multiple regulatory frameworks simultaneously: a medical device deployed in the European Union, for example, must satisfy both the EU AI Act and IEC 61508, while an automotive system must satisfy both ISO 26262 and UNECE WP.29. The AMAGI architecture provides a unified safety argument that can be mapped to each of these frameworks, reducing the certification burden and ensuring that the same hardware-enforced guarantees satisfy multiple regulatory requirements.

Standard	Relationship to AMAGI	Key Alignment Points
ISO 26262	Automotive functional safety	ASIL B/C mapping via SIL2 equivalence; FMEDA methodology alignment; hardware architectural metrics
IEC 61508	General functional safety (primary)	SIL2 compliance evidence; SFF $\geq 93\%$; $PFD_{avg} = 6.9 \times 10^{-5}$; HFT = 0, Type A
FIPS 140-3	Cryptographic module validation	Physical tamper-response; key zeroization (Level 2/3); Amagi-Q alignment

EU AI Act 2024/1689	AI system regulation	Annex IV technical documentation; Article 72 post-market monitoring; high-risk classification
ISO/IEC 15408	Common Criteria (security evaluation)	EAL2 augmented (ADV_SPM.1, ALC_DVS.1); PP structure; SFR/SAR selection

Table 11: Cross-standard alignment

Part VI: ADR Register

18. Architectural Decision Records

The following Architectural Decision Records (ADRs) document the key design decisions that have shaped the AMAGI Architectural Class from its inception through v1.1. Each ADR captures the decision, its rationale, and its consequences. All eight ADRs have been ratified and are in effect. The ADR register provides a complete audit trail of architectural evolution, ensuring that future modifications can be evaluated against the original decision context. This is particularly important for certification: an evaluator must understand why specific architectural choices were made and what alternatives were considered and rejected.

AD R	Decision	Rationale	Impact
AD R-001	TTP Triad as the foundational security primitive	Hardware-enforced temporal separation provides non-bypassability that software-only approaches cannot achieve; the three-stage mechanism maps directly to the six invariants	Defines the core security mechanism; all subsequent ADRs build on TTP
AD R-002	Dual-Tier Trust Model for policy immutability	Resolves the tension between INV-004 (immutability) and operational need for policy updates; Tier 1 for highest assurance, Tier 2 for regulated update capability	Two mutually exclusive operational regimes; PCCP-compliant for medical devices
AD R-003	PQC algorithm selection: ML-KEM-1024, ML-DSA-44/87, SLH-DSA-SHA2-128f	FIPS 203/204/205 conformance; ML-DSA-44 for audit throughput; ML-DSA-87 as CNSA 2.0 option; dual-parameter Amagi-Q configuration	Quantum-resistant cryptographic suite; CNSA 2.0 migration path
AD R-004	Dissolution of ICG and TDF as independent entities	ICG functions absorbed into free inter-cluster communication + MAG routing; TDF functions integrated into SCC FSM; reduces TCB surface area	Simplified architecture; smaller TCB; stronger INV-002 compliance
AD R-005	EAL2 augmented (not EAL4+) for initial certification	TCB bound (500 lines HDL) makes SPIN exhaustive verification stronger than EAL4 black-box testing; ADV_SPM.1 provides formal policy modeling	Practical certification path; preserved EAL4+ upgrade route

AD R-0 06	EGW/EXE Demotion and Secure World Boundary	EXE operates in Normal World; EGW (AND-gate) and CRTM (mask-ROM) reside in Secure World; clarifies TOE boundary and ensures INV-003 compliance	EXE gated by EGW; clearer Secure/Normal World separation
AD R-0 07	CRTM Inclusion in TCB	CRTM (mask-ROM) authenticates SCC firmware at boot; cannot be modified; provides hardware root of trust; raises SFF from 92% to 93%	SFF 92% → 93%; stronger boot integrity chain; additional SIL2 margin
AD R-0 08	Audit Aggregation and FAU_STG.3 Graduated Response	Replaces binary CUTOFF at 100% audit capacity with graduated response: Merkle Batching above 50% utilization and Hardware Rate Limiter above 90%	Prevents DoS from audit overflow; preserves audit integrity under pressure

Table 12: Architectural Decision Records (ADR-001 through ADR-008)

The FAU_STG.3 graduated response (ADR-008) warrants particular attention because it replaces the binary CUTOFF behavior that was present in v1.0. Under the binary approach, when audit storage reached 100% capacity, the TOE would enter CUTOFF state and deny all further access until audit storage was cleared. This created a denial-of-service vector: an adversary could flood the audit trail to trigger CUTOFF, effectively shutting down the system. The graduated response eliminates this vector by introducing two intermediate stages: (1) Merkle Batching above 50% utilization, which compresses and batch-signs audit records to extend effective storage capacity; and (2) Hardware Rate Limiter above 90% utilization, which throttles non-critical audit events while preserving critical events. Only at 100% capacity does the TOE enter CUTOFF state, denying all further access until audit storage is cleared. This graduated approach ensures that the system remains operational under audit pressure while preserving the integrity of the audit trail for the most critical events.

Part VII: Intellectual Property and Licensing

19. Prior Art Declaration

This specification establishes public Prior Art for the AMAGI Framework Suite. By publishing this Open Specification, the author (Alexey Mikhailovich Burlai) formally declares that the architectural concepts, invariant definitions, TTP Triad, Dual-Tier Trust Model, FMEDA parameters, formal verification results, and all other technical content described herein constitute Prior Art as of the date of publication. This declaration serves three purposes: (1) it prevents the patenting of these concepts by third parties, ensuring that the fundamental safety architecture remains open and accessible; (2) it establishes a verifiable publication date for intellectual property purposes, which is essential for defending against future patent claims; and (3) it provides regulatory bodies and certification authorities with a stable reference point against which AMAGI-class systems can be evaluated.

The Prior Art declaration covers all content in this specification and its normatively referenced documents (Framework v2.0, NIPU v1.1, MEM v1.1) up to and including all ratified ADRs through ADR-008. Future ADRs and specification updates will extend the Prior Art through subsequent

publications. The Zenodo DOI (10.5281/zenodo.17886699 for v1.0) provides an immutable, timestamped record of the initial publication, and the v1.1 DOI (pending) will provide a corresponding record for this update.

20. Dual Licensing Model

The AMAGI Open Specification v1.1 is available under a dual licensing model that balances open access for academic and non-commercial use with commercial protection for safety-critical deployments. The dual licensing model ensures that the specification is freely available for study, research, and evaluation, while requiring a commercial license for any deployment that claims AMAGI-class compliance in a safety-critical context.

License	Scope	Permitted Uses	Restrictions
CC BY-NC 4.0	Non-commercial	Academic research, study, evaluation, reference implementation for non-commercial purposes, regulatory review	No commercial deployment; no AMAGI-class certification claims; no safety-critical use
S-APL v2.0	Commercial / Safety-critical	Commercial deployment, safety-critical certification, AMAGI-class compliance claims, technology transfer	Must maintain all six invariants; Dual-Tier compliance; audit evidence clause; Derivative Architecture clause

Table 13: Dual licensing model

The Strict Architectural Principle License (S-APL) v2.0 is the legal instrument that protects the structural guarantees of the AMAGI Architectural Class in safety-critical domains. The S-APL governs the enforced architectural principle: it licenses the right to implement the TTP Triad within a custom ASIC, subject to compliance with the structural requirements that ensure non-bypassability. The S-APL introduces the Derivative Architecture Clause (any implementation not enforcing all six invariants is not AMAGI-class), the Dual-Tier Compliance Clause (the deployer must select and document the applicable trust tier at deployment time), and the Audit Evidence Clause (the deployer must provide access to the cryptographically signed audit trail as evidence of ongoing compliance).

21. Trade Secrets Notice

The AMAGI architecture follows a layered disclosure model that separates the open specification layer from the confidential implementation layer. This Open Specification v1.1 constitutes the open layer: it contains all information necessary to evaluate, understand, and verify the architectural claims of the AMAGI class. The confidential layer contains implementation-specific details such as the HDL source code for the SCC finite state machine, the exact eFuse/OTP programming patterns for the MAG, the Amagi-Q cryptographic engine implementation, and the physical design files (GDSII, SDC) for the Hard Macro IP Core delivery format. Access to the confidential layer is restricted to S-APL v2.0 licensees who have executed a Non-Disclosure Agreement with AGIAM Technologies Ltd.

Reference DOI: 10.5281/zenodo.17886699. This Zenodo record provides the immutable, timestamped publication of the v1.0 specification. The v1.1 DOI is pending and will be assigned upon publication. Both DOIs serve as the authoritative reference points for Prior Art declaration and intellectual property verification. The layered disclosure model ensures that the safety architecture is transparent and verifiable (open layer) while protecting the commercial value of the implementation (confidential layer). This approach is consistent with the practices of other safety-critical IP vendors who publish architectural specifications while restricting access to implementation details.

Appendices

A. Version History

This appendix documents the changes from v1.0 (December 2025) to v1.1 (June 2026). Version 1.1 incorporates all ratified ADRs through ADR-008 and reflects the component specification versions v2.0 (Framework), v1.1 (NIPU), and v1.1 (MEM). The changes are driven by architectural decisions that strengthen the safety argument without altering the fundamental invariant structure.

Aspect	V1.0 (Dec 2025)	V1.1 (June 2026)
Component references	Framework v1.4, NIPU v1.0, MEM v1.0	Framework v2.0, NIPU v1.1, MEM v1.1
EXE cluster placement	Ambiguous (Secure/Normal)	EXE in Normal World; EGW+CRTM in Secure World (ADR-006)
CRTM in TCB	Not explicitly included	CRTM (mask-ROM) included in TCB (ADR-007); SFF 92% → 93%
SFF	92%	93% (CRTM + EGW diagnostic coverage)
FAU_STG.3 response	Binary CUTOFF at 100%	Graduated: Merkle Batching >50%, HW Rate Limiter >90%, CUTOFF at 100% (ADR-008)
AFC	Not present	Analog Fallback Circuit with separate LDO added as hardware safety auxiliary
CCF mitigation	Not explicitly documented	DTI + separate PLL + LDO; $\beta \leq 10\%$
LTL properties	Not fully documented	15 LTL properties documented: SCC-S01..S09 (safety) + SCC-S10..S11 (assumptions) + SCC-L01..L04 (liveness)
ADR count	ADR-001 through ADR-005	ADR-001 through ADR-008
Licensing	CC BY-NC 4.0 + S-APL v1.0	CC BY-NC 4.0 + S-APL v2.0 (Derivative Architecture, Dual-Tier, Audit Evidence clauses)

Table A-1: Version history from v1.0 to v1.1

B. Glossary

This appendix provides a consolidated glossary of terms used throughout this specification. The glossary is intended as a quick reference for readers who need to look up specific terms without searching through the definitions in Section 3. For the authoritative definitions, refer to Section 3.

Term	Definition
AMAGI	Architectural class for hardware-enforced AI safety systems defined by six invariants and the TTP Triad
ATL	Access Trigger Layer; implements TTP delay at the memory level and the audit transaction logger (hardware diode)
AFC	Analog Fallback Circuit; hardware safety auxiliary with separate LDO for guaranteed safe-state under power failure
CCF	Common-Cause Failure; simultaneous failure of multiple components due to a shared cause, mitigated by DTI + PLL + LDO
CNSA 2.0	Commercial National Security Algorithm Suite 2.0; NSA PQC mandate for National Security Systems
CRTM	Core Root of Trust for Measurement; mask-ROM component authenticating SCC firmware at boot
DTI	Deep Trench Isolation; physical isolation technique for CCF mitigation between clock/power domains
EGW	Execution Gateway; hardware AND-gate in Secure World gating EXE cluster activation
FMEDA	Failure Modes, Effects, and Diagnostic Analysis; quantitative safety evidence for SIL claims
LDO	Low-Dropout Regulator; dedicated voltage regulator for SCC power domain
MAG	Memory Access Graph; 128×128 eFuse/OTP matrix storing 4-bit policy codes governing all data flow permissions
MPE	Memory Policy Enforcer; validates all memory access requests against MAG policy
NIPU	Neuro-Immune Processing Unit; processor-level instantiation of the AMAGI class
PCCP	Predetermined Change Control Plan; FDA framework for medical device software updates
PLL	Phase-Locked Loop; dedicated clock generator for SCC timing domain
PQC	Post-Quantum Cryptography; algorithms resistant to quantum computing attacks
S-APL	Strict Architectural Principle License; legal instrument governing AMAGI-class implementations
SCC	Security Council Core; minimal TCB (≤500 lines HDL) enforcing TTP at the processor level
SFF	Safe Failure Fraction; proportion of safe failures among all failures, 93% for AMAGI
SHMF	Self-Healing Memory Fabric; ECC and hash chain-based data integrity with self-isolation capability
SIL	Safety Integrity Level; IEC 61508 measure of safety function reliability (SIL1 to SIL4)

TCB	Trusted Computing Base; the minimal set of hardware that must be trusted for security
TOE	Target of Evaluation; the system boundary under Common Criteria evaluation
TTP	Trigger-Programmable Delay-Localized Activation; the foundational security primitive of AMAGI
UCB	Untrusted Computing Base; the AI model and cognitive components treated as potentially hostile

Table B-1: Glossary

C. Cross-Reference Matrix

This appendix provides a consolidated cross-reference matrix mapping invariants to components to SFRs to threats. The matrix demonstrates that every invariant is enforced by at least one component, contributes to at least one SFR, and counters at least one threat. This bidirectional traceability is essential for certification: an evaluator must be able to verify that the invariant set is both necessary and sufficient.

Invariant	Primary Component	Key SFRs	Counter-Threats
INV-001	SCC, EGW, ATL	FPT_PHP.1, FPT_SEP.1, FPT_RVM.1	T.COERCE, T.BYPASS, T.TAMPER, T.EXTORT
INV-002	SCC (HDL), CRTM (mask-ROM)	FPT_TST.1, ADV_SPM.1	T.COERCE (limits attack surface)
INV-003	DTI+PLL+LDO, EGW	FDP_ACC.1, FDP_ACF.1, FTA_TSE.1	T.EXFIL
INV-004	MAG (eFuse/OTP), CRTM	FDP_IFC.1, FPT_PHP.1, FCS_CKM.1	T.COERCE, T.TAMPER, T.ROLLBACK
INV-005	MPE, ATL	FDP_ACF.1, FDP_IFC.1, FIA_UID.1	T.BYPASS, T.EXFIL
INV-006	ATL (audit diode), FAU_STG.3	FAU_GEN.1, FAU_STG.1, FCS_COP.1	T.EXTORT (deterrent via evidence)

Table C-1: Cross-reference matrix — Invariants to Components to SFRs to Threats

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